Markov State Modelling for the folding of HP35 villin headpiece

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In this work, I compare different feature sets and dimensionality reduction methods for the construction of Markov state models (MSMs) of the folding of the HP35 villin headpiece.

I. INTRODUCTION

A. Brief description of the biological problem, knowns and unknowns

Markov State Models (MSMs) have become a standard framework for extracting long-timescale kinetics from molecular dynamics simulations and for constructing coarse-grained descriptions of biomolecular conformational landscapes [7]. Building an MSM typically involves several stages, including feature selection, dimensionality reduction, clustering, transition matrix estimation, validation and coarse-graining. In recent years, MSM methodology has evolved rapidly, with advances in state construction, uncertainty quantification, and the incorporation of machine learning for identifying collective variables and building models at scale, such as VAMPnets [9] and time-lagged autoencoders [10]. Beyond methodological developments, MSMs are increasingly integrated with experimental observables and applied to larger and more complex systems. Together, these efforts reflect a field that continues to evolve in both rigor and scope [1].

B. Defining the question

The purpose of this university project is to move the first steps into the world of Markov state modeling. The goal is to review and benchmark the classical MSM construction pipeline, with particular emphasis on feature selection and dimensionality reduction stages. Establishing a clear baseline based on conventional methods is a necessary step before evaluating more sophisticated machine-learning approaches in future studies.

C. Defining the plan: translating the question on the language of the computer (brief)

As a case study, I consider the villin headpiece subdomain HP35, a small fast-folding protein for which extensive molecular dynamics trajectories are available. I compare two commonly used feature representations, native-contact distances and backbone dihedral angles, combined with two dimensionality-reduction methods, principal component analysis (PCA) and time-lagged independent component analysis (tICA). The resulting four MSMs are evaluated in terms of (i) their ability to reproduce the folding kinetics of the system and (ii) their ability to provide an interpretable characterization of the metastable conformational states.

II. METHODS

A. Dataset assembly

All analyses presented in this work are based on a single $300\,\mu$s molecular dynamics trajectory of the fast-folding Lys24-Nle/Lys29-Nle mutant of the 35-residue villin headpiece subdomain (HP35), simulated at 360 K. HP35 is a prototypical ultrafast-folding protein with an experimentally measured folding time of approximately $0.7,\mu$s at this temperature and has become a standard benchmark system for Markov state model construction and validation [4–6]. The trajectory was originally generated by D. E. Shaw Research on the Anton supercomputer and reported by Piana *et al.* [2]. Their simulations predicted a folding time of approximately $1.5,\mu$s, reasonably close to the experimental estimate of $0.73,\mu$s for the same mutant, although still larger by a factor of about $\times 2.5$. At the time of this work, the original trajectory generated by D. E. Shaw Research was temporarily unavailable due to maintenance of the data repository. I therefore used a preprocessed version distributed by Nagel *et al.* through their GitHub repository, which has been adopted in recent benchmark studies of Markov state modeling [4, 5]. The timeseries consists of approximately 1.5×10^6 samples, with a sampling interval of $\simeq 200\,ps$.

B. Methods used for analysis (in case you developed a method, it goes into Results)

Markov state models Ire built using the Deeptime library [8]. Since MSM construction is a multi-step process SF1, it is especially important to understand clearly the choices made at each step and validate them before moving on to the next one. The principle "garbage in, garbage out" applies here, and a mistake at an early stage can propagate and affect the final results. For this reason, in this section I describe synthetically the motivation for each step and define the methods that Ire investigated

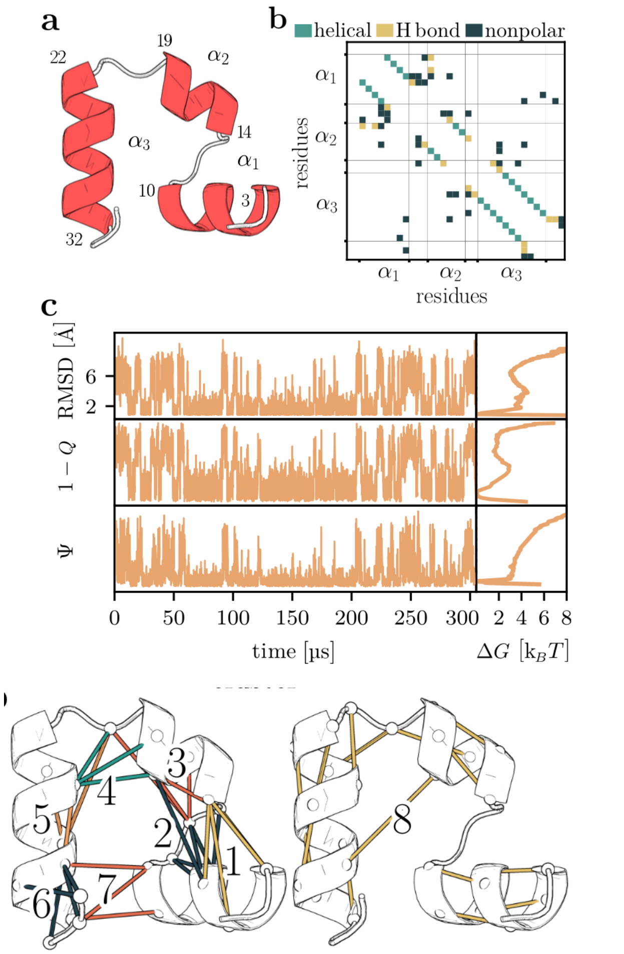

FIG. 1: *Folding of villin headpiece (HP35). (a) Molecular structure, consisting of three α-helices (residues $3-10$, $14-19$ and $22-32$) that are connected by two loops. (b) Contact map showing the 42 native contacts of HP35. (c) Time evolution of the RMSD, compared to $1-Q$ (with Q being the fraction of native contacts) and Φ representing the sum over the backbone dihedral angles ϕ_i of the three α-helices. The right side shows the corresponding free energy curves (in units of k_BT) along these coordinates.*

and compared in this work.

Feature definition and filtering The feature representations employed in this work Ire adopted from the preprocessing pipeline of Nagel *et al.* [4], to which I refer the reader for a detailed description. Two feature sets Ire considered: native-contact distances and backbone dihedral angles. Native contacts Ire defined from folded conformations observed in the simulation and filtered using distance and frequency criteria, yielding 42 contact-distance features. Backbone conformations Ire represented by (ϕ) and (ψ) dihedral angles using the angular transformation proposed by Sittel *et al.* [13] to account for periodicity. To remove Iakly correlated motions unlikely to contribute to the folding dynamics, the MoSAIC filtering procedure of Nagel *et al.* [4] was applied. The final dihedral feature set consisted of 62 coordinates, with (ψ_1), (ψ_2), (ψ_{34}), (ϕ_2), (ϕ_{34}), and (ϕ_{35}) excluded.

Dimensionality reduction and clustering Dimensionality reduction has the scope of reducing the dimensionality of the features for the subsequent definition of the states of the markov chain (referred to as microstates). The feature representations described above are high-dimensional and therefore unsuitable for direct clustering. A dimensionality-reduction step is commonly employed to obtain a compact representation of the conformational ensemble while retaining the information relevant to the slow dynamics. Ideally, conformations that rapidly interconvert should remain close in the reduced space, whereas conformations separated by slow transitions should be Ill separated. Since microstates are subsequently defined by clustering in this reduced space, the quality of the dimensionality reduction directly affects the quality of the resulting Markov state model. Many methods have been proposed, both linear and nonlinear, some aided by deep learning. In this work I restrict ourselves to considering two widely used linear dimensionality-reduction methods: principal component analysis (PCA) and time-lagged independent component analysis (tICA). PCA identifies orthogonal directions of maximal variance in the data, whereas tICA identifies directions of maximal autocorrelation at a specified lag time. Because slow molecular processes are typically associated with slowly decorrelating coordinates, tICA is often regarded as more suitable for MSM construction. I compare different variations of tICA, namely i) no normalization, ii) kinetic map rescaling and ii) compute time rescaling [S1]. After dimensionality reduction, conformations Ire partitioned into microstates using the k-means clustering algorithm. **MSM estimation and hyperparameter selection with VAMP** I estimated the transition matrix from the empirical count matrix of transitions betIen microstates and applying a maximum likelihood estimator. The selection of the lag time was one of the hyperparameters to optimize and its optimization was done as explained in the next section.

The number of retained components and the number of microstates constitute important hyperparameters of the MSM construction procedure.

For tICA, the relevant hyperparameters are the number of retained time-lagged independent components and the tICA lag time (τ_{tICA}). The corresponding eigenvalues quantify the autocorrelation of the projected coordinates at lag time (τ_{tICA}).

Historically, the selection of these hyperparameters relied largely on heuristic criteria and subsequent validation of the resulting MSM through Chapman-Kolmogorov (CK) tests. More recently, variational approaches such as the Variational Approach for Markov Processes (VAMP) [11, 12] have introduced objective scoring functions that can be used to compare different choices of features, dimensionality-reduction parameters, and clustering schemes. These scores provide a principled framework for model selection and are discussed in more detail in the Supporting Information.

The hyperparameters of the dimensionality-reduction and clustering pipeline Ire selected by comparing cross-validated VAMP-2 scores over a grid of tICA lag times, retained tICs, and numbers of k-means microstates. The model with the highest validation VAMP-2 score was selected, subject to subsequent MSM validation. Maybe also implied timescales test?

MSM validation The models selected Ire validated with a Chapman-Kolmogorov test [S2].

C. Statistical analysis

Here talk about everything you do after you have the validated MSM: the TPT analysis, the estimation of pathways, and the coarse graining.

III. RESULTS

This is the real talk, the biological talk! Comparative analysis of the results of the various MSM constructed.

Sections describing different aspects of your analysis (3a-3d). All sections should be organised as follows, the text is continuous, broken only to a few paragraphs in each section:

- Question studied in the section and the goal of this calculation.
- Few sentences on how it was done, mostly referring to datasets and approaches in the Methods.
- Actual results and observations, each supported by either a i) table or ii) figure or iii) number in the text.
- You need to interpret the observation. For example, do not only write X ¿ Y, but also the meaning of this relationship with respect to your question.
- When presenting the results, refer to the respective Figure and Table, where it was shown.
- You can add more sentences for the interpretation. Please use precise phrases for the level of evidence, if the results 'suggest', 'indicate', 'show','demonstrate', 'support', 'corroborate'. Do not put too much 'hypothesis', and 'propose' these are synthetic phrases.
- End each section with a conclusion sentence for example, 'Taken together the results consistently show that X is larger than Y, indicating that X was an ancestor of Y'.

IV. DISCUSSION

Here you try to put your results into a broader context and present your scientific view/model based on that. This is a continuous text, broken into paragraphs. In case you have two distinct 'huge' conclusions, you may generate separate sections (max 2). Chapter 4 should be organised as follows:

- You may phrase your biological question again.
- Briefly summarise the results (max one paragraph). You do not repeat the words you used earlier, make it concise. Make it evident how your results ansIr the question you have asked.
- Compare your results with something in the literature (use PubMed, or literature indicated in UniProt, PDB or any datasets I learnt about). 1-2 articles enough. The point is that you
- show how your results relate to previous ones. Different scenarios: 'support', 'in contrast', 'in addition', 'gives insights into'.
- If possible, present a model. This can be a molecular 'mechanism', 'scenario', 'reason'. For example: 'A and B belong to the same family'; 'mutation XD likely changes the interaction in liver'; 'in disease T I have less phosphorylation sites or the level of phosphorylation decreased'; 'in this experiment, there is a problem with degradation of protein P. ' etc.
- The model can be presented as: 'propose', 'suggest', 'indicate','hypothesize', or even 'support' or 'demonstrate'. If possible try to present a cartoon figure or scheme.

V. CONCLUSIONS

One paragraph, emphasizing the main finding and formulate a message for the future (perspective), how to continue from here, what kind of further studies are needed. You may have Conclusion as chapter 5.

VI. REFERENCES

The text should contain references, to previous results, data, methods, softwares, etc.. Format is unrestricted,

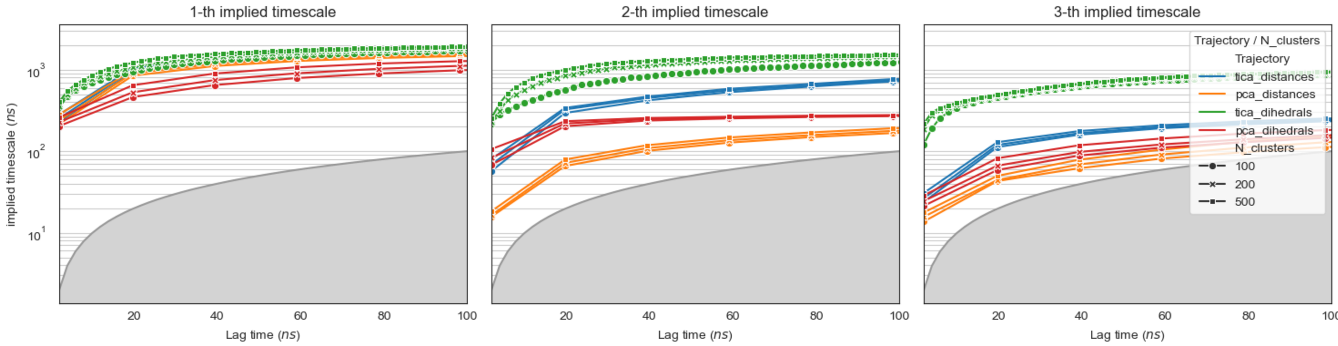

FIG. 2: **Implied timescales convergence test.**

but you should use these refs in the text, and list them at the end. I expect ca 15-20 refs listed.

Data availability statement: Preprocessed trajectories of the HP35 villin headpiece are kindly made available by Nagel et al. and downloadeable at their GitHub repository https://github.com/moldyn/HP35-DESRES. All code necessary to replicate the analysis shoId in this work is available on GitHub at https://github.com/miriamzara/MSM.

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SUPPORTING INFORMATION

S1. Dimensionality reduction and clustering

PCA and tICA are both linear dimensionality reduction techniques, i.e. they project the data onto a linear subspace of the original feature space. Let $\{\mathbf{x}(t)\}_{t=0}^{T-1}$ be a multidimensional, discrete time-series, where each snapshot is a column vector, of dimension D, corresponding to an arbitrary, vectorized representation of a protein conformation (e.g. a set of dihedral angles). Suppose the data has been sampled from a *stationary* and *time-reversible* stochastic process. Without loss of generality, I assume that the data have been centered, $\mathbb{E}[\mathbf{x}] = 0$. Using the braket notation $\langle x| := \mathbf{x}^T, |x\rangle := \mathbf{x}$, I define the covariance matrix:

$$\mathbf{C}_0 := \mathbb{E}[|x\rangle \langle x|] \tag{1}$$

and the time-lagged covariance matrix:

$$\mathbf{C}_{0,\tau} := \mathbb{E}\left[|x(t)\rangle \langle x(t+\tau)|\right]. \tag{2}$$

For a stationary reversible process,

$$\mathbf{C}_{0,\tau} = \mathbf{C}_{\tau,0}^T = \mathbf{C}_{\tau,0},$$

so that the lagged covariance matrix may be denoted simply by $\mathbf{C}_\tau$.

If $|\hat{u}\rangle \langle \hat{u}|$ is a projection operator (i.e. $|\hat{u}\rangle$ is a unit vector), then:

$$\langle \hat{u}| \mathbf{C}_0 |\hat{u}\rangle = \mathbb{E}[\langle \hat{u}|x\rangle \langle x|\hat{u}\rangle] = \mathbb{E}[\langle \hat{u}|x\rangle^2] \tag{3}$$

is the variance of the projection of the data along $|\hat{u}\rangle$, and

$$\begin{aligned}\mathrm{ACF}_{\hat{u}}(\tau) &= \frac{\langle \hat{u}| \mathbf{C}_\tau |\hat{u}\rangle}{\langle \hat{u}| \mathbf{C}_0 |\hat{u}\rangle} \\ &= \frac{\mathbb{E}[\langle \hat{u}|x(t)\rangle \langle \hat{u}|x(t+\tau)\rangle]}{\mathbb{E}[\langle \hat{u}|x\rangle^2]}\end{aligned} \tag{4}$$

its autocorrelation at lag time τ.

PCA seeks the direction of maximal variance. The first principal component is defined as

$$\begin{aligned}|\hat{u}_1\rangle = &\arg\max \langle \hat{u}| \mathbf{C}_0 |\hat{u}\rangle \\ &\text{subject to } \|\hat{u}\| = 1\end{aligned} \tag{5}$$

this optimization problem can be solved introducing a Lagrange multiplier λ, and the stationarity condition yields

$$\mathbf{C}_0 |\hat{u}_i\rangle = \lambda_i |\hat{u}_i\rangle, \tag{6}$$

i.e. the projection versor is an eigenvector of the covariance matrix. The corresponding eigenvalue equals the variance of the data projected onto the principal component:

$$\langle \hat{u}_i| \mathbf{C}_0 |\hat{u}_i\rangle = \lambda_i \cdot \langle \hat{u}_i|\hat{u}_i\rangle = \lambda_i.$$

Then the second principal component is defined as the direction of maximal variance orthogonal to the first, giving the optimization problem:

$$\begin{aligned}|\hat{u}_2\rangle = &\text{argmax} \langle u| \mathbf{C}_0 |u\rangle \\ &\text{subject to } \langle u|u\rangle = 1 \\ &\text{and } \langle u|\hat{u}_1\rangle = 0\end{aligned} \tag{7}$$

which again can be solved with Lagrange multipliers. Iterating, this implies that the principal components form an orthonormal basis of eigenvectors of the covariance matrix, i.e. they diagonalize the covariance matrix. Dimensionality reduction ($d < D$) is achieved by taking the components along the first d principal components $\{|\hat{u}_1\rangle, \dots, |\hat{u}_d\rangle\}$.

$$z_i = \langle \hat{u}_i|x\rangle, \qquad \mathbf{z} = U^T\mathbf{x}. \tag{8}$$

This is a linear projection onto an ortonormal basis:

$$|x\rangle \to |z\rangle := \sum_{i=1}^{d} \langle x|\hat{u}_i\rangle \, |\hat{u}_i\rangle$$

By construction, in the reduced space components are uncorrelated and ordered by decreasing variance:

$$\mathbb{E}[|z\rangle \langle z|] = \mathrm{diag}(\lambda_1, \cdots \lambda_d) \tag{9}$$

tICA Unlike PCA, tICA [15] seeks directions $|u\rangle$ that maximize the autocorrelation Eq.(4) at a fixed lag time τ. The lagtime τ is a free parameter. The autocorrelation is invariant under scaling of the projection vector $|u\rangle$. Differently from PCA, the normalization constraint is usually implemented as $\langle u| \mathbf{C}_0 |u\rangle = 1$, as this choice simplifies the optimization problem. Formally, tICA solves the optimization problem:

$$\begin{aligned}|\hat{u}_1\rangle = &\text{argmax} \langle u| \mathbf{C}_\tau |u\rangle \\ &\text{subject to } \langle u| \mathbf{C}_0 |u\rangle = 1\end{aligned} \tag{10}$$

using the same procedure as for PCA, I can solve this with Lagrange multipliers, and show that

$$\mathbf{C}_\tau |\hat{u}_1\rangle = \lambda_1 \mathbf{C}_0 |\hat{u}_1\rangle , \tag{11}$$

.

The generalized eigenvalue λ_i is equal to the autocorrelation of the projected coordinate $\langle \hat{u}_i | x(t) \rangle$ at lag time τ. Assuming that this coordinate approximates a single relaxation mode of the dynamics, its autocorrelation decays exponentially, yielding

$$\lambda_i \approx e^{-\tau/t_i}, \tag{12}$$

where t_i is the corresponding relaxation timescale. In practice, the relation is approximate because a tIC may contain contributions from multiple dynamical modes.

Similarly as done with PCA, I can iterate this procedure and define the second tIC as the direction of maximal autocorrelation at lag time τ that is *uncorrelated* with the first:

$$\begin{aligned} |\hat{u}_2\rangle = & \text{argmax} \langle u| \mathbf{C}_\tau |u\rangle \\ & \text{subject to } \langle u| \mathbf{C}_0 |u\rangle = 1 \\ & \text{and } \langle u| \mathbf{C}_0 |\hat{u}_1\rangle = 0 \end{aligned} \tag{13}$$

the solution $|\hat{u}_1\rangle$ is again a solution of the same generalized eigenvalue problem as in Eq.(10). generalized eigenvalue problem. Dimensionality reduction ($d < D$) is achieved by taking the components of the data $\mathbf{x}(t)$ along the first d tICs $\{|\hat{u}_1\rangle , \ldots , |\hat{u}_d\rangle\}$.

$$\mathbf{z}(t) = (\langle x(t)|\hat{u}_i\rangle , \cdots , \langle x(t)|\hat{u}_d\rangle) \tag{14}$$

By construction, in the reduced space components z_i have unit variance, are uncorrelated at zero lag, and are ordered by decreasing autocorrelation at lag time τ.

$$\mathbb{E}[|z\rangle \langle z|] = \mathbb{I} \tag{15}$$

tICA as a diagonalization in the whitened coordinate space It is worth noting that unlike in PCA, the tIC vectors are generally not orthogonal under the standard Euclidean inner product. Instead, as I said, they satisfy the generalized orthogonality relation

$$\langle u_i| \mathbf{C}_0 |u_j\rangle = \delta_{ij}.$$

i.e. they corresponding components z_i and z_j are *uncorrelated* at zero lag. Nevertheless, the tICA procedure can be reformulated as a standard eigenvalue problem in a transformed coordinate system. The key idea is to first apply a whitening transformation, which rescales and rotates the original data so that all directions have unit variance and are mutually uncorrelated. Define the ZCA whitening transformation

$$|y\rangle = \mathbf{C}_0^{-1/2} |x\rangle$$

where $C_0^{-1/2}$ is symmetric and positive definite. By construction, the covariance matrix in the transformed space is the identity:

$$\mathbb{E}\left[|y\rangle \langle y|\right] = \mathbf{C}_0^{-1/2} \mathbf{C}_0 \mathbf{C}_0^{-1/2} = \mathbf{I}.$$

and time-lagged covariance matrix in the whitened space is

$$\hat{\mathbf{C}}_\tau = \mathbf{C}_0^{-1/2} \mathbf{C}_\tau \mathbf{C}_0^{-1/2}$$

$\hat{\mathbf{C}}_\tau$ is symmetric and can thus be diagonalized:,

$$\hat{\mathbf{C}}_\tau |v_i\rangle = \lambda_i |v_i\rangle , \qquad \langle v_i|v_j\rangle = \delta_{ij}.$$

If $|u_i\rangle$ solves the tICA problem (10), then $|v_i\rangle = \mathbf{C}_0^{1/2} |u_i\rangle$ is an eigenvector of $\hat{\mathbf{C}}_\tau$ with the same eigenvalue.

Indeed, multiplying the generalized eigenvalue problem (10) from the left by $\mathbf{C}_0^{-1/2}$ and defining $|v_i\rangle = \mathbf{C}_0^{1/2} |u_i\rangle$ yields

$$\hat{\mathbf{C}}_\tau |v_i\rangle = \lambda_i |v_i\rangle .$$

Consequently, the tICA components z_i correspond to the ortogonal projection of the whitened data $\mathbf{y}$ onto the ortonormal basis defined by $\{|v_i\rangle\}$:

$$z_i(t) = \langle x(t)|u_i\rangle = \langle x(t)|C_0^{-1/2} v_i\rangle = \langle y(t)|v_i\rangle .$$

Alternative generalized eigenvector normalizations for tICA
The construction presented above corresponds to the original formulation of tICA, in which the covariance matrix of the transformed data is the identity (15). HoIver, alternative rescaling of the transformed coordinates have been used in the literature and are usually implemented in softwares to endow Euclidean distances in the reduced space with a more direct kinetic interpretation. These can be particularly useful when the tICA coordinates are subsequently clustered to construct a Markov state model, as I do in this work.

The most commonly used choice is the *kinetic map* in which each tIC component z_i is multiplied by its corresponding eigenvalue,

$$\tilde{z}_i(t) = \lambda_i z_i(t). \qquad (16)$$

With this normalization, the coordinates representing the sloIst processes "Iight more", and a unit step in those directions in the reduced space is assigned a larger distance. An alternative is the *commute map*, where each coordinate is rescaled according to its implied timescale,

$$\tilde{z}_i(t) = \sqrt{\frac{t_i}{2}} z_i(t), \qquad t_i = -\frac{\tau}{\ln \lambda_i}. \qquad (17)$$

With this normalization, Euclidean distances approximate commute distances, i.e. the expected time required to travel from one configuration to another and back. This is a variation on the same idea behind the kinetic map, being eigenvalues and timescales connected by (12).

S2. Estimation and validationof Markov state models

CK test

To insert.

S3. VAMP scores

S4. Transition path theory

S5. PCCA and MSM coarse-graining

SUPPLEMENTARY FIGURES

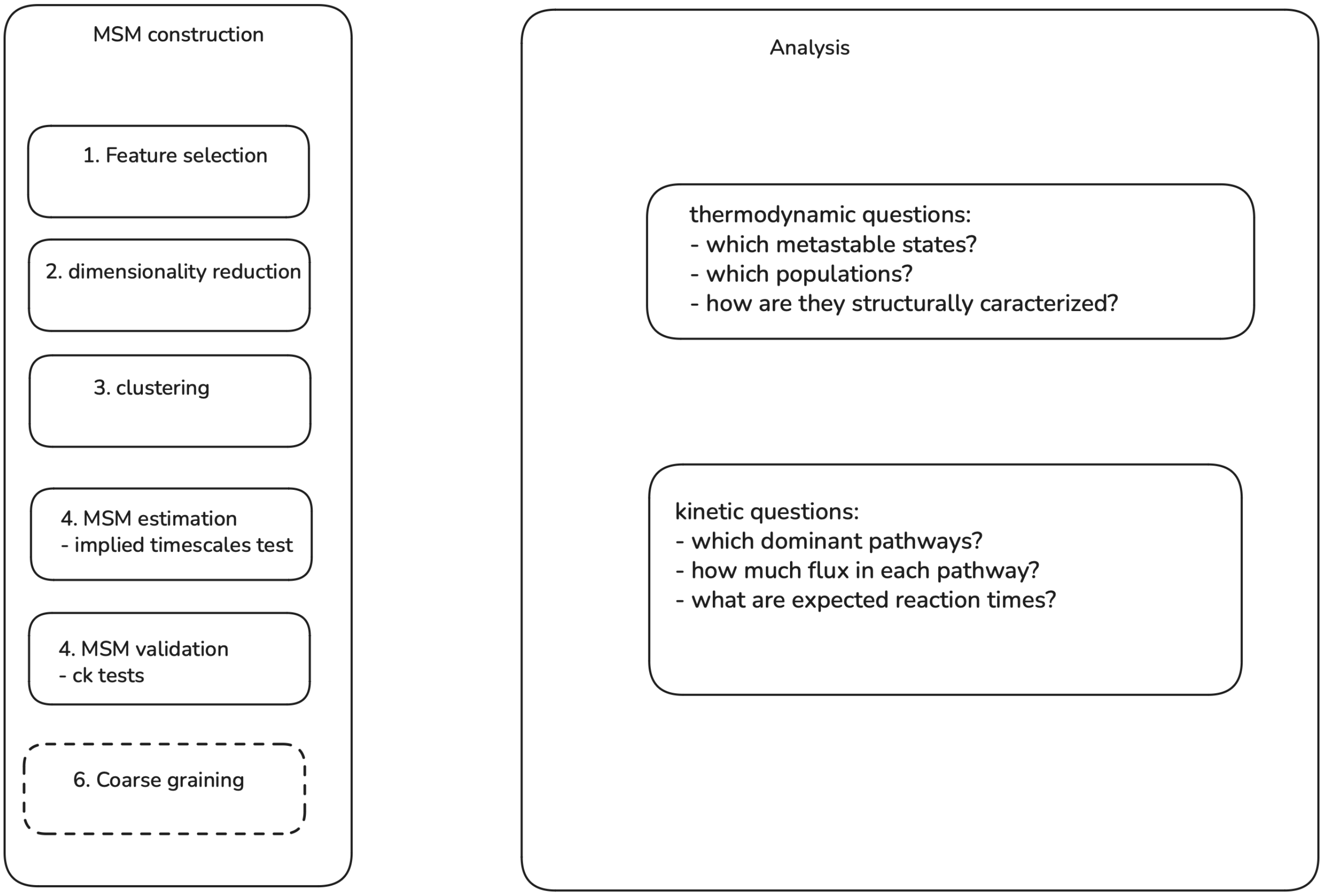

FIG. SF1: **Flowchart of the MSM construction pipeline used in this work**.

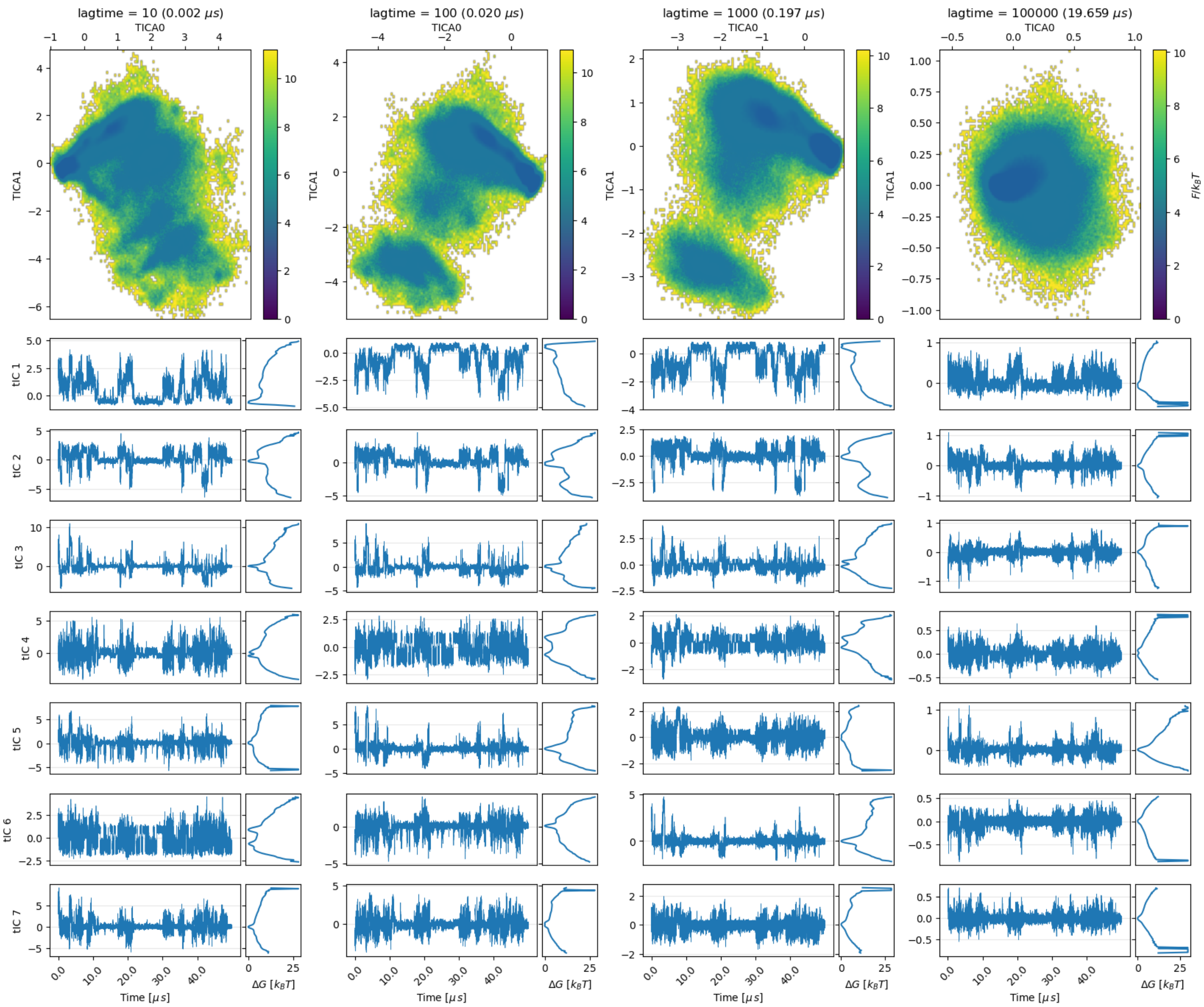

FIG. SF2: **tICA projections obtained from the dihedral-angle feature set using kinetic-map scaling**. Columns correspond to different tICA lag times spanning four orders of magnitude ($10, 10^2, 10^3, 10^5$ frames; values in μs are reported above each panel). The top row shows the free-energy landscape projected onto the first two tICs, while the loIr panels report the first seven tIC time series (left) and their one-dimensional free-energy profiles (right) for the first $40\,\mu s$ of simulation. A useful tICA projection should separate slowly interconverting conformations into distinct regions of configuration space and produce components exhibiting metastable, jump-like dynamics. Very short lag times lead to projections dominated by fast fluctuations, whereas excessively large lag times wash out the dynamical structure and yield nearly unimodal projections. Intermediate lag times ($\tau \sim 1{-}2\,\mu s$) provide the clearest separation of metastable states and the most pronounced residence/jump behaviour in the leading components. Based on this analysis, a lag time of approximately $2\,\mu s$ and the first five tICs Ire retained for subsequent MSM construction.

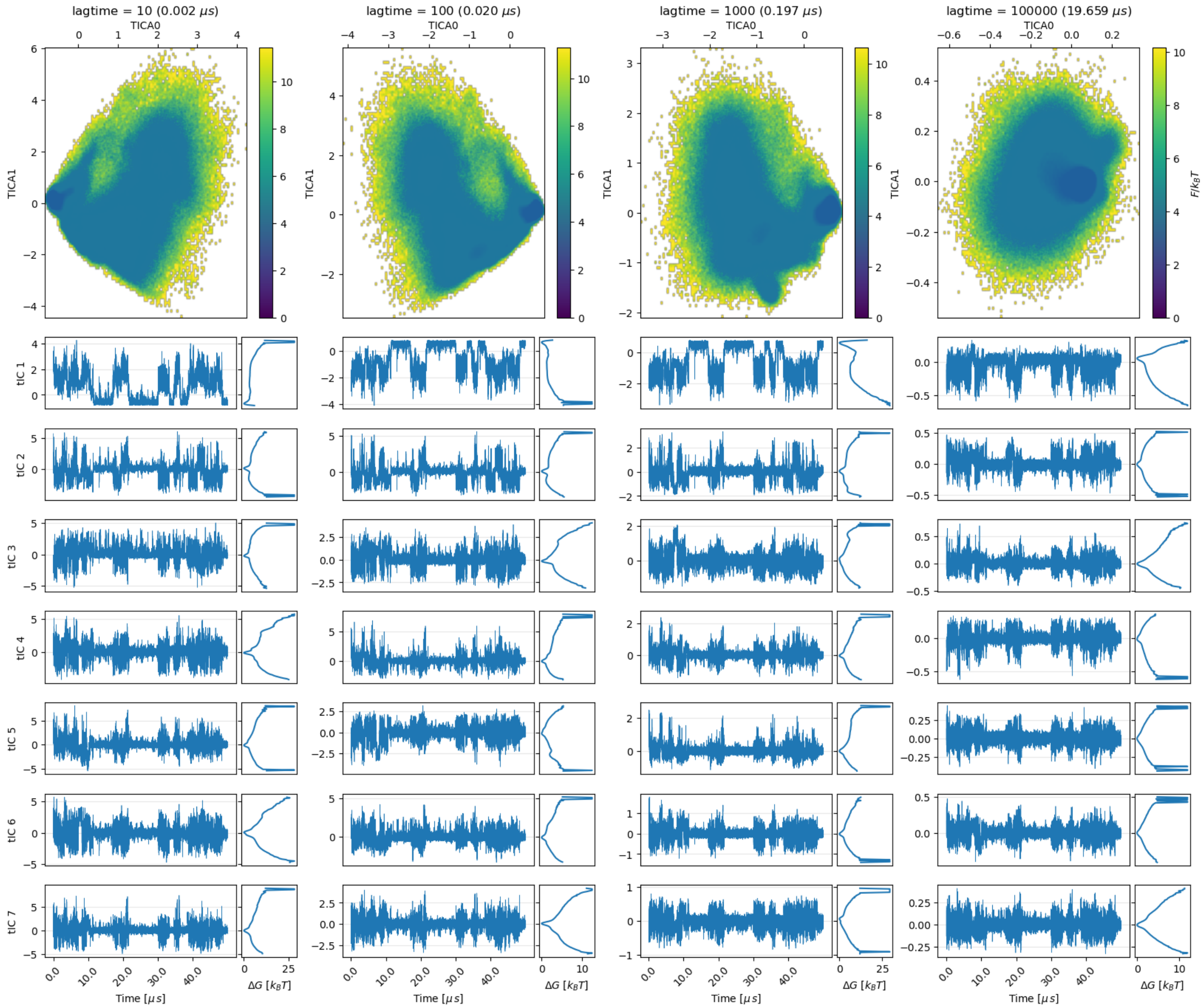

FIG. SF3: **tICA projections obtained from the contact-distance feature set using kinetic-map scaling**. Compared to the dihedral-angle representation, the contact-distance features yield tICA coordinates with less clearly separated free-energy minima and less pronounced metastable, jump-like dynamics. This suggests a Iaker separation of slow conformational processes in the resulting low-dimensional projection. A lag time of $\tau = 10^3 frames (\approx 0.2\,\mu s)$ was selected to maintain consistency with the analysis performed on the dihedral feature set. At this lag time, the free-energy profiles become largely unimodal beyond the first few tICs, indicating that most of the slow dynamical information is concentrated in the leading components. Although the first three tICs appear to capture the dominant slow processes, as a more conservative choice the first five components Ire retained for subsequent MSM construction.

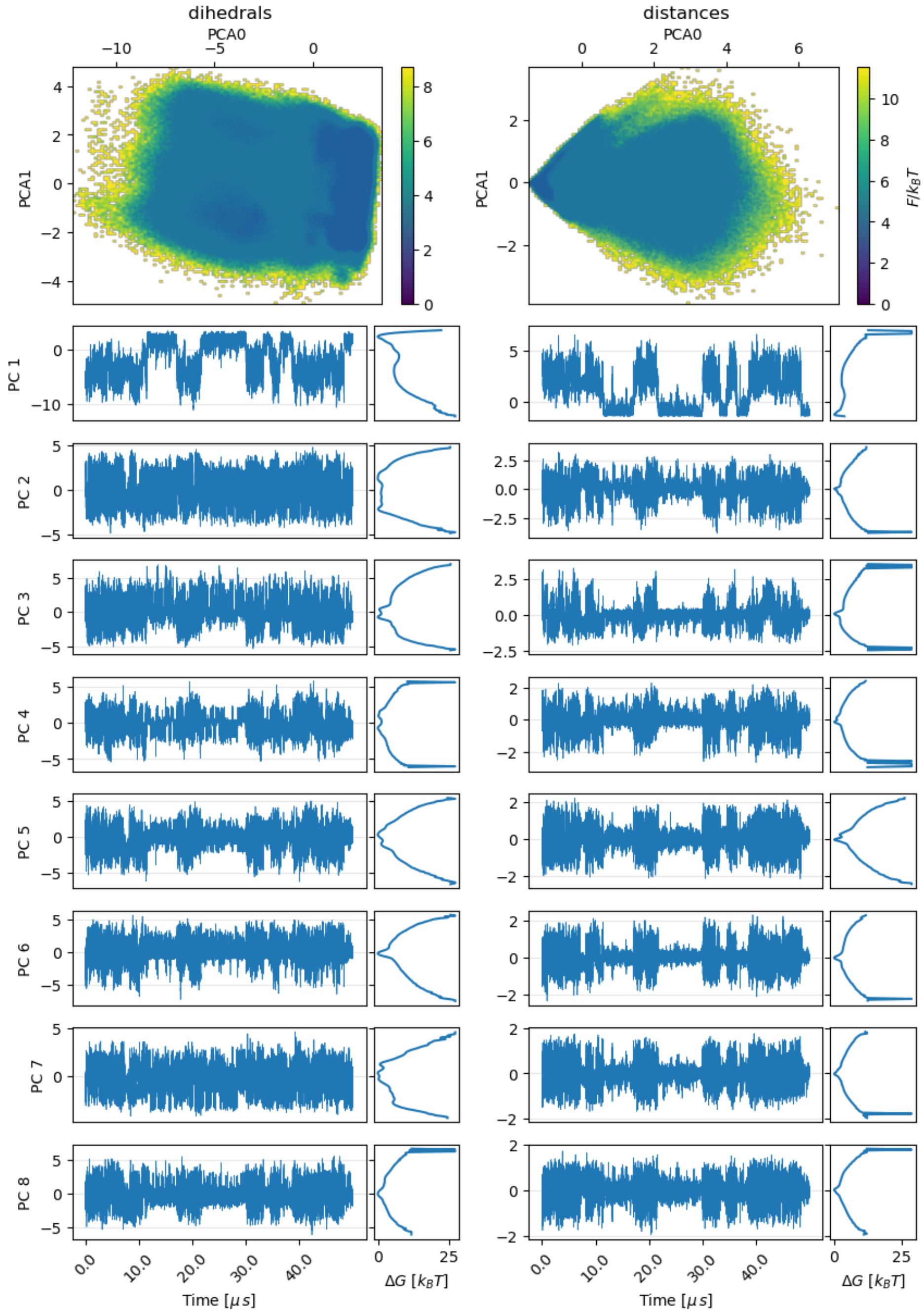

FIG. SF4: **Comparison of PCA projections obtained from the contact-distance feature set and from the dihedrals feature set.**. In contrast to tICA, the PCA projections provide a less clear separation of metastable conformational states. Free-energy profiles exhibit Iaker clustering, the one-dimensional free-energy profiles are predominantly unimodal, and the principal-component time series do not display a pronounced residence/jump behavior, with exception of the first PC. Consistent with the tICA results, the dihedral-angle representation yields a more structured low-dimensional projection than the contact-distance features. For subsequent MSM construction I retain the first 7 components for the dihedrals set and the first 5 for the contact-distance set.